

genes_visualization

Import required library

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

```
dataset_path <- "/Users/sorenbasnet/Documents/Github/STAT_443_FINAL_PROJECT/data/raw/c11_broad_2022"
```

```
data_mutations <- read.table(file.path(dataset_path, "data_mutations.txt"), header=TRUE, sep="\t")
```

```
colnames(data_mutations)
```

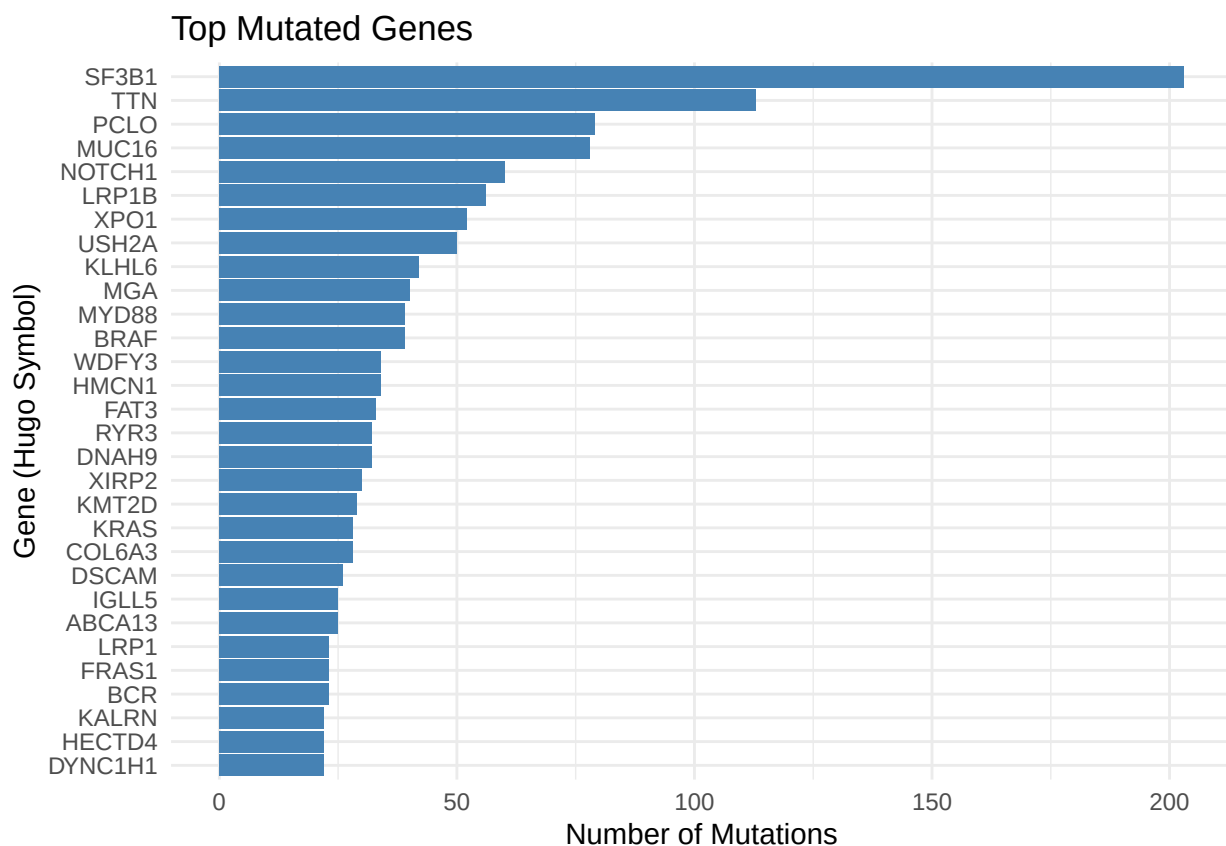
```
## [1] "Hugo_Symbol"           "Entrez_Gene_Id"  
## [3] "Center"                "NCBI_Build"  
## [5] "Chromosome"           "Start_Position"  
## [7] "End_Position"         "Strand"  
## [9] "Consequence"          "Variant_Classification"  
## [11] "Variant_Type"         "Reference_Allele"  
## [13] "Tumor_Seq_Allele1"    "Tumor_Seq_Allele2"  
## [15] "dbSNP_RS"             "dbSNP_Val_Status"  
## [17] "Tumor_Sample_Barcode" "Matched_Norm_Sample_Barcode"  
## [19] "Match_Norm_Seq_Allele1" "Match_Norm_Seq_Allele2"  
## [21] "Tumor_Validation_Allele1" "Tumor_Validation_Allele2"  
## [23] "Match_Norm_Validation_Allele1" "Match_Norm_Validation_Allele2"  
## [25] "Verification_Status"  "Validation_Status"  
## [27] "Mutation_Status"      "Sequencing_Phase"  
## [29] "Sequence_Source"      "Validation_Method"  
## [31] "Score"                "BAM_File"  
## [33] "Sequencer"            "t_ref_count"  
## [35] "t_alt_count"          "n_ref_count"  
## [37] "n_alt_count"          "HGVSc"  
## [39] "HGVSp"                "HGVSp_Short"  
## [41] "Transcript_ID"        "RefSeq"  
## [43] "Protein_position"     "Codons"  
## [45] "Exon_Number"          "HGVSp_Short.1"  
## [47] "Annotation_Status"
```

```
dim(data_mutations)
```

```
## [1] 20004      47

top_genes <- data_mutations %>%
  count(Hugo_Symbol, sort = TRUE) %>%
  head(30)

ggplot(top_genes, aes(x = reorder(Hugo_Symbol, n), y = n)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  coord_flip() +
  labs(
    title = "Top Mutated Genes",
    x = "Gene (Hugo Symbol)",
    y = "Number of Mutations"
  ) +
  theme_minimal()
```



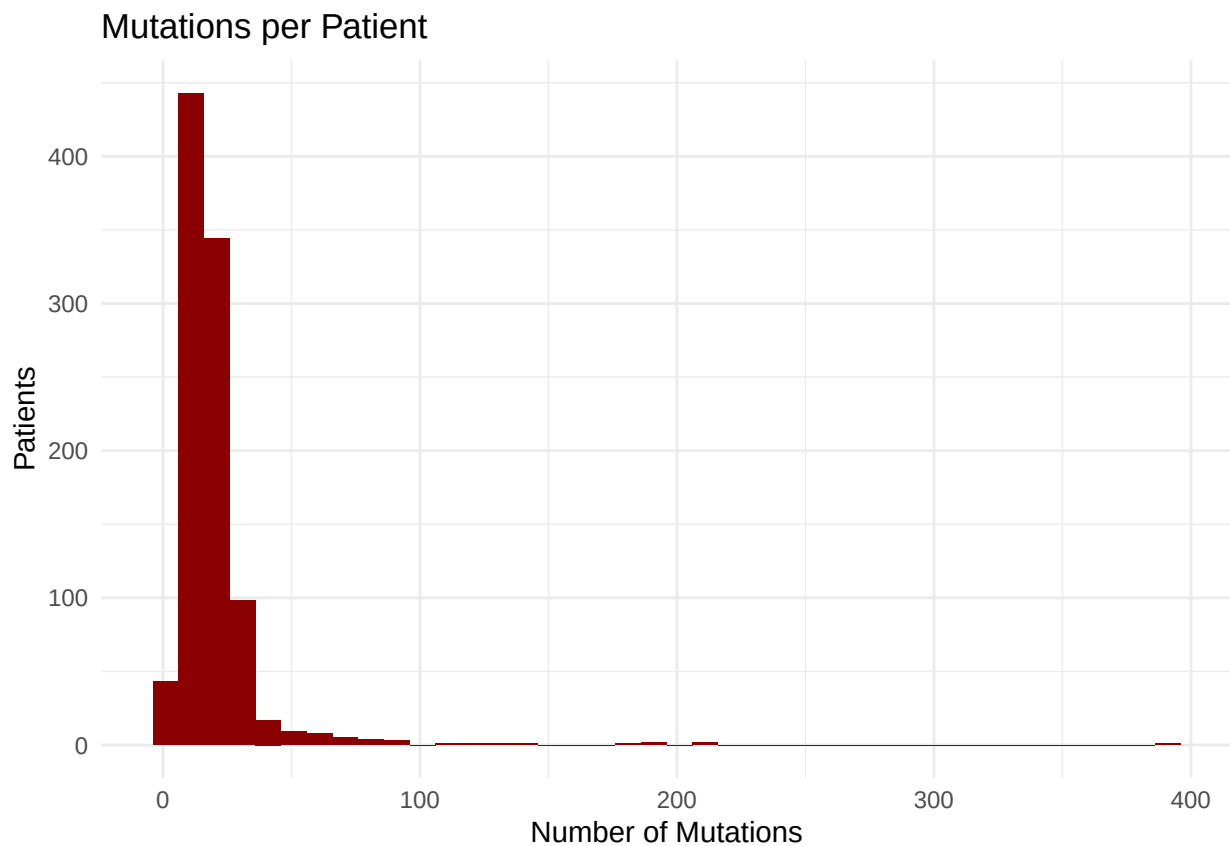
In a dataset of mutations with ~20000 genes, it is important to note how rare these genes are in the dataset. This may/may not influence results.

Mutation Per Person

```
patient_mutations <- data_mutations %>%
  count(Tumor_Sample_Barcode)

ggplot(patient_mutations, aes(x = n)) +
  geom_histogram(bins = 40, fill = "darkred") +
  labs(
    title = "Mutations per Patient",
    x = "Number of Mutations",
  )
```

```
y = "Patients"
) +
theme_minimal()
```



Gene $\times$ Patient Heatmap

```
library(tidyr)

gene_patient <- data_mutations %>%
  select(Tumor_Sample_Barcode, Hugo_Symbol) %>%
  distinct() %>%
  mutate(value = 1) %>%
  pivot_wider(
    names_from = Hugo_Symbol,
    values_from = value,
    values_fill = 0
  )
```

Now convert to matrix and plot:

```
#library(pheatmap)

#mat <- as.matrix(gene_patient[, -1])
#rownames(mat) <- gene_patient$Tumor_Sample_Barcode

#pheatmap(mat)
```

Mutation along gene

```
ggplot(data_mutations, aes(x = Protein_position)) +  
  geom_histogram(bins = 50) +  
  theme_minimal() +  
  labs(title = "Mutation Position Distribution")
```

```
## Warning: Removed 122 rows containing non-finite outside the scale range  
## (`stat_bin()`).
```

